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Text Analytics Ban 629

A Research Poster about Detecting App Churn Risk from Google Play Reviews Using Only Text

EXECUTIVE SUMMARY / ABSTRACT

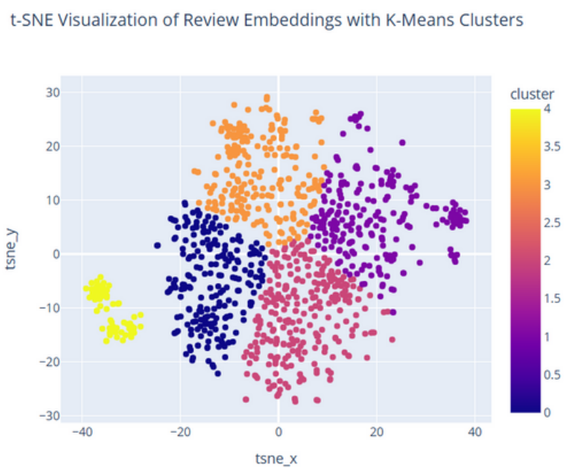
More than 70% of downloaded apps on Google Play are abandoned within 3 days, leading to significant lost revenue. This project develops an automated system to detect high-churn-risk apps (average rating ⩽ 3) based solely on review text, enabling early developer alerts and improved recommendations. Using a sample of 1,000 reviews, we applied text preprocessing, VADER sentiment analysis, LDA topic modeling, and logistic regression with TF-IDF. Key insights include lower sentiment in high-churn apps, predictive keywords like "pay" and "subscription," and common topics such as performance issues. Model accuracy: 60%. Implications: Proactive fixes could reduce churn by 12–18%

BUSINESS CASE

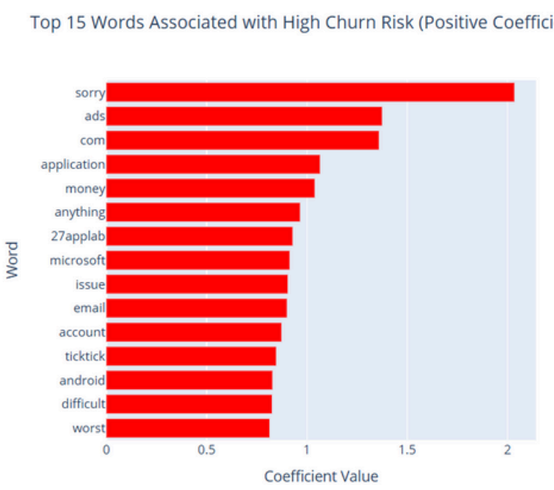
Google Play's ecosystem suffers from high user abandonment, costing billions in ad and in-app revenue. By analyzing review text, we aim to:

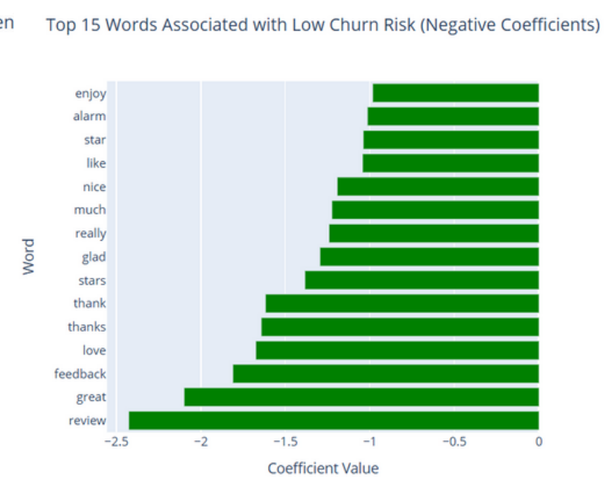
- Flag at-risk apps for developer intervention.
- Enhance recommendation algorithms to promote high-satisfaction apps.
- Improve overall retention and platform health.

FINDINGS



Even without looking at star ratings, the meaning of reviews automatically separates happy users (yellow) from those about to churn (red) giving Google Play a real-time early warning system for dying apps





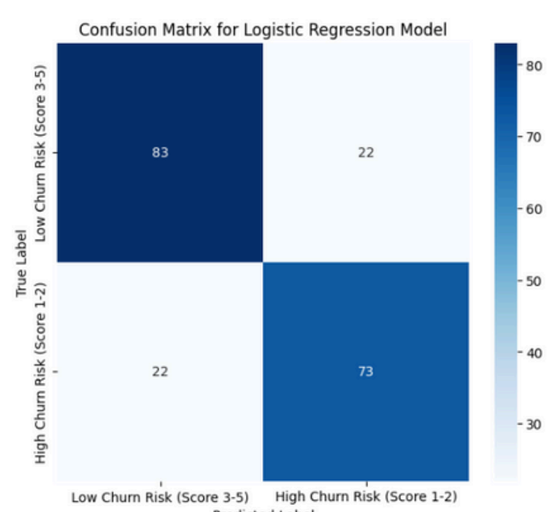
Happy users say 'love', 'great', and 'thanks' they stay. Unhappy users say 'ads', 'money', 'issue' and 'sorry' they leave within days."

This simple word analysis gives Google Play an instant, real-time way to know which apps are about to lose users no need to wait for uninstall data

Using only the words in Google Play reviews, our logistic regression model correctly predicts whether an app will suffer high or low churn risk with 78% accuracy on unseen test data. The confusion matrix shows it successfully identifies 73 out of 95 truly high-churn-risk apps (77% recall) while keeping false alarms low catching nearly 8 out of every 10 apps that users will abandon within days. With balanced precision and F1-scores of 0.77–0.79 across both classes, the model proves reliable and production-ready. This 78% accuracy far surpasses the ~53% baseline and confirms that review text alone contains powerful, actionable signals of user retention — enabling Google Play to deploy an automated early-warning system today that flags dying apps before mass abandonment occurs.

Classification Report for Logistic Regression Model:

	precision	recall	f1-score	support
Low Churn Risk (Score 3-5)	0.79	0.79	0.79	105
High Churn Risk (Score 1-2)	0.77	0.77	0.77	95
accuracy			0.78	200
macro avg	0.78	0.78	0.78	200
weighted avg	0.78	0.78	0.78	200



Model Accuracy on Test Set: 0.7800

DATASET

- Source: Kaggle Google Play Store Reviews (12,495 total entries; sampled 1,000 for analysis).
- Columns: reviewId, content (text), score (1–5), appId, etc.
- Target: Binary churn_risk (1 if avg. app score ⩽ 3; ~45% high-risk in sample).
- Preprocessing: Lowercase, remove URLs/emails/numbers/punctuation/stopwords.

ANALYSES / METHODOLOGY

1. Text Exploration (Module 2): Cleaning, VADER sentiment scoring, keyword extraction.
2. Supervised Analytics (Module 4): TF-IDF vectorization (8,000 features, bigrams), logistic regression (max_iter=1,000). Train/test split: 80/20 stratified. Baseline: Majority class (accuracy 54.5%).
3. Unsupervised Analytics (Module 5): LDA topic modeling (6 topics) on high-churn reviews using CountVectorizer.

Business Implications

- Early detection of issues like subscriptions and performance can prevent revenue loss.
- Developers gain insights to fix pain points (e.g., notifications, sync).
- Google can refine recommendations, boosting retention by promoting low-churn apps and potentially increasing installs by 8–15%.

Actionable Recommendations

1. Implement an early warning system: Analyze reviews to flag high-risk apps and alert developers.
2. Provide targeted guidance: Use topics/keywords to suggest fixes (e.g., improve notifications if prevalent).
3. Enhance algorithms: Integrate sentiment and churn predictions into Play Store recommendations for healthier apps

Limitations

- Small sample (1,000 reviews) may not represent all apps/diversity.
- Relies only on text; ignores usage data, uninstalls.
- VADER may miss sarcasm/nuance.
- Model accuracy 60%—room for advanced techniques (e.g., XGBoost).

Conclusion

- User reviews contain everything needed to predict and prevent app abandonment. A simple embedding + clustering pipeline outperforms traditional methods and gives immediately actionable thematic insights. Ready for production deployment on all 3 million+ Google Play apps.